Codificación de fuentes de información

Teoría de la Información y la Comunicación

Codificación de fuentes de información

C A D B C C B A C A B C C D B

A	B	C	D	Total
3	4	6	2	**15**

Codificación de fuentes de información

C A D B C C B A C A B C C D B

A	B	C	D	Total
0.20	0.27	0.40	0.13	1

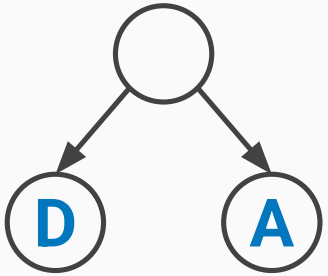
Algoritmo de Huffman

A	0.20	
B	0.27	
C	0.40	
D	0.13	

Algoritmo de Huffman

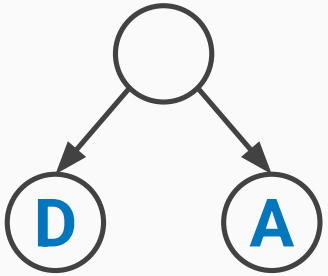
C	0.40	
B	0.27	
A	0.20	
D	0.13	

Algoritmo de Huffman



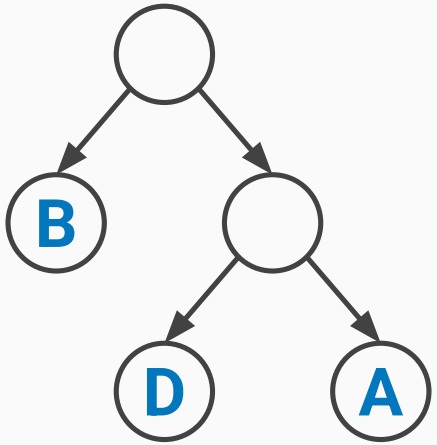
C	0.40	
B	0.27	
A	0.33	
D		

Algoritmo de Huffman



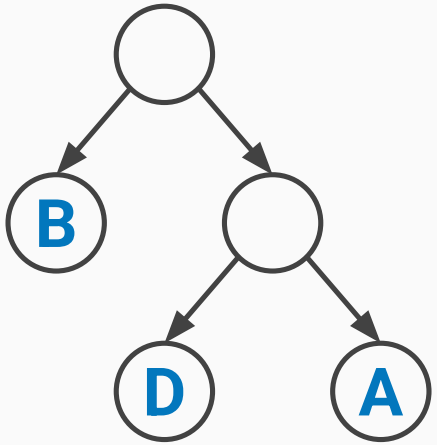
C	0.40	
A	0.33	
D		
B	0.27	

Algoritmo de Huffman



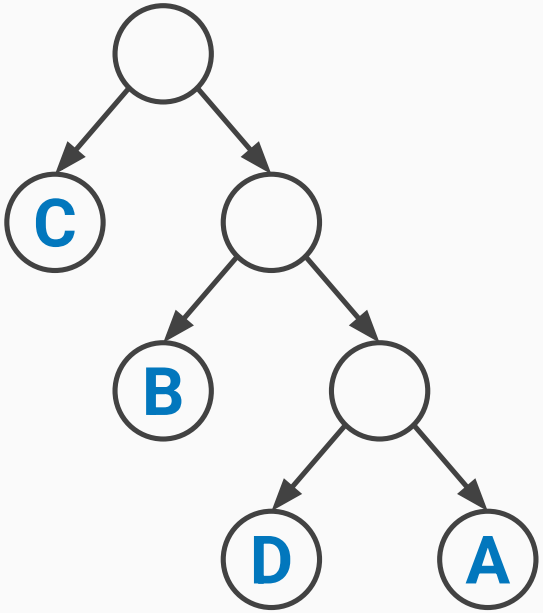
C	0.40	
A	0.60	
B		
D		

Algoritmo de Huffman



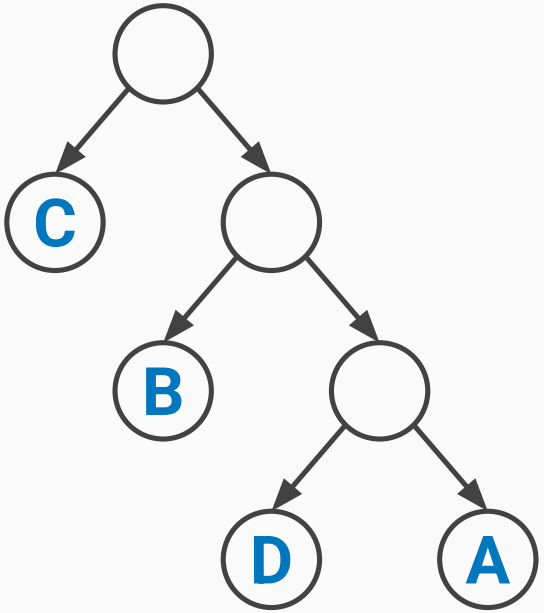
A	0.60	
B		
D		
C	0.40	

Algoritmo de Huffman



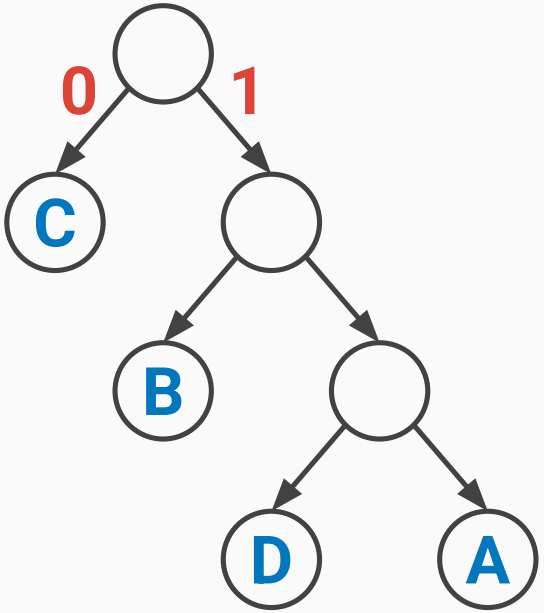
A	1.00	
B		
C		
D		

Algoritmo de Huffman



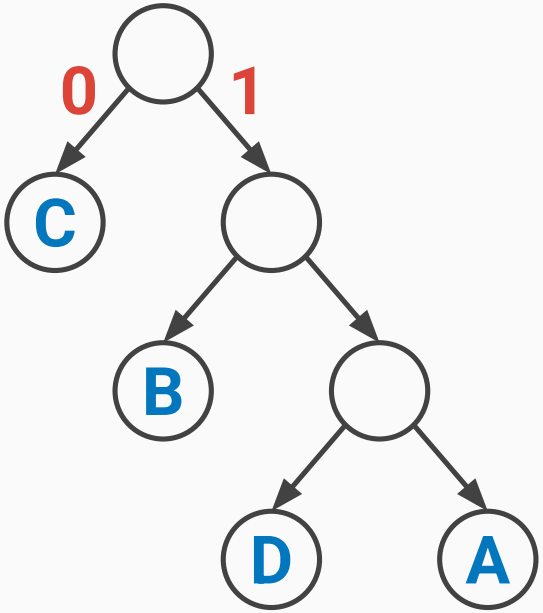
A	0.60	
B		
D		
C	0.40	

Algoritmo de Huffman



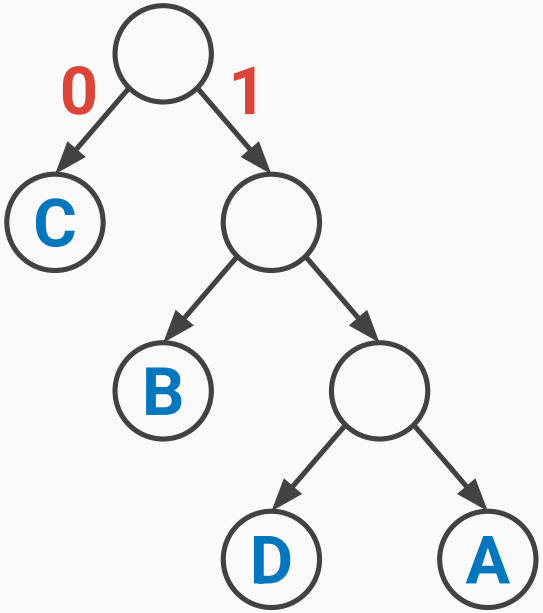
A	0.60	1
B		
D		
C	0.40	0

Algoritmo de Huffman



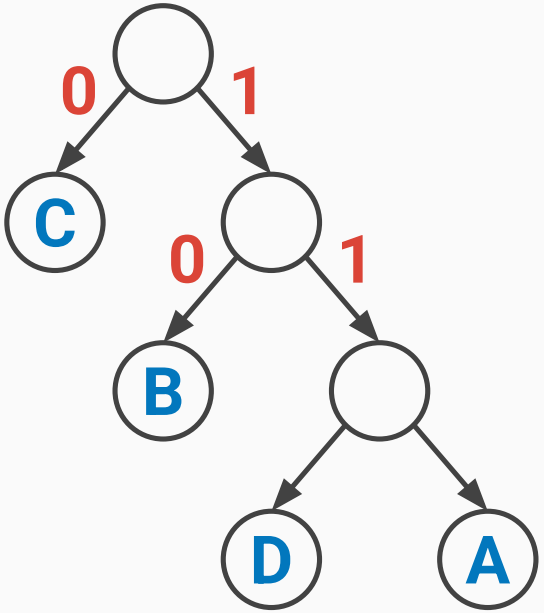
C	0.40	0
A	0.33	
D		
B	0.27	

Algoritmo de Huffman



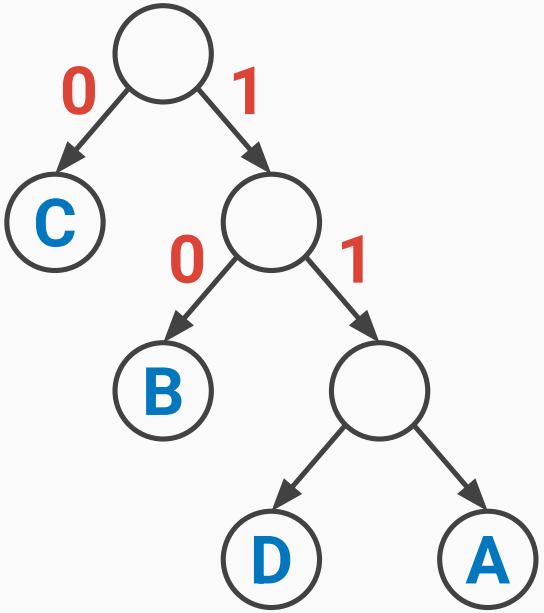
C	0.40	0
A	0.33	1
D		
B	0.27	1

Algoritmo de Huffman



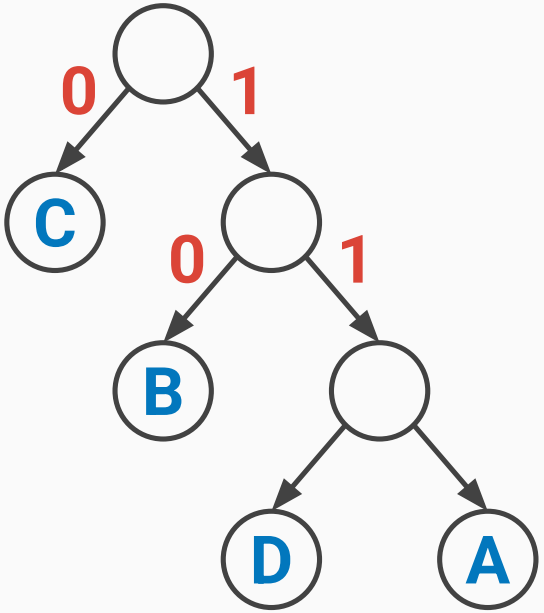
C	0.40	0
A	0.33	11
D		
B	0.27	10

Algoritmo de Huffman



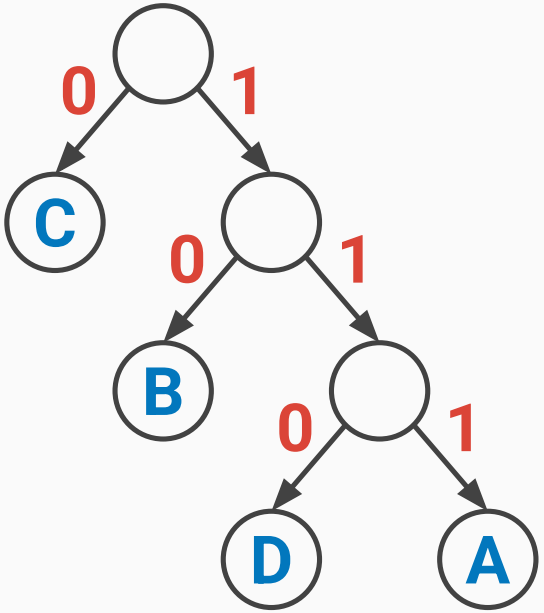
C	0.40	0
B	0.27	10
A	0.20	
D	0.13	

Algoritmo de Huffman



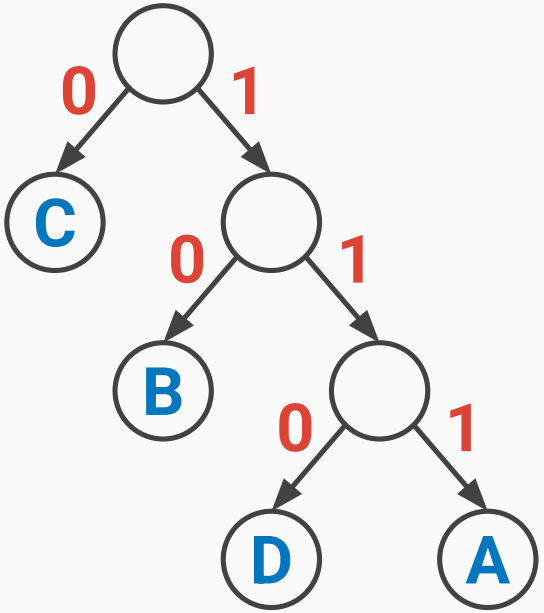
C	0.40	0
B	0.27	10
A	0.20	11
D	0.13	11

Algoritmo de Huffman



C	0.40	0
B	0.27	10
A	0.20	111
D	0.13	110

Algoritmo de Huffman



A	0.20	111
B	0.27	10
C	0.40	0
D	0.13	110

Algoritmo de Huffman

A	0.20	111
B	0.27	10
C	0.40	0
D	0.13	110

$H(S) = 1.89$ bits

$L = 1.93$ bits

$\eta = 0.9772$

$R = 0.0228$

Algoritmo de Huffman

C A D B C C B A C A B C C D B

0 111 110 10 0 0 10 111 0 111 10 0 0 110 10

A	0.20	111
B	0.27	10
C	0.40	0
D	0.13	110

Algoritmo de Shannon-Fano

A	0.20
B	0.27
C	0.40
D	0.13

A	0.20	
B	0.27	
C	0.40	
D	0.13	

Algoritmo de Shannon-Fano

C	0.40
B	0.27
A	0.20
D	0.13

C	0.40	
B	0.27	
A	0.20	
D	0.13	

Algoritmo de Shannon-Fano

C	0.40	0.40
B	0.27	
A	0.20	0.33
D	0.13	

C	0.40	
B	0.27	
A	0.20	
D	0.13	

Algoritmo de Shannon-Fano

C	0.40	0.40
B	0.27	0.60
A	0.20	
D	0.13	

C	0.40	
B	0.27	
A	0.20	
D	0.13	

Algoritmo de Shannon-Fano

C	0.40	1
B	0.27	
A	0.20	0
D	0.13	

C	0.40	
B	0.27	
A	0.20	
D	0.13	

Algoritmo de Shannon-Fano

C	0.40	1
B	0.27	
A	0.20	0
D	0.13	

C	0.40	1
B	0.27	0
A	0.20	0
D	0.13	0

Algoritmo de Shannon-Fano

C	0.40	1
B	0.27	
A	0.20	0
D	0.13	

C	0.40	1
B	0.27	0
A	0.20	0
D	0.13	0

Algoritmo de Shannon-Fano

C	0.40	1	
B	0.27		
A	0.20	0	0.60
D	0.13		

C	0.40	1
B	0.27	0
A	0.20	0
D	0.13	0

Algoritmo de Shannon-Fano

C	0.40	1	
B	0.27		0.27
A	0.20	0	
D	0.13		0.13

C	0.40	1
B	0.27	0
A	0.20	0
D	0.13	0

Algoritmo de Shannon-Fano

C	0.40	1	
B	0.27		0.27
A	0.20	0	0.33
D	0.13		

C	0.40	1
B	0.27	0
A	0.20	0
D	0.13	0

Algoritmo de Shannon-Fano

C	0.40	1	
B	0.27	0	1
A	0.20		0
D	0.13		

C	0.40	1
B	0.27	0
A	0.20	0
D	0.13	0

Algoritmo de Shannon-Fano

C	0.40	1	
B	0.27	0	1
A	0.20		0
D	0.13		

C	0.40	1
B	0.27	01
A	0.20	00
D	0.13	00

Algoritmo de Shannon-Fano

C	0.40	1	
B	0.27	0	1
A	0.20		0
D	0.13		

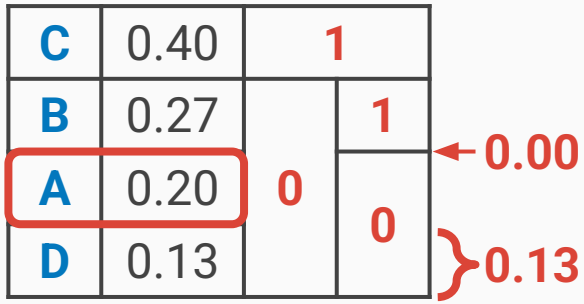
C	0.40	1
B	0.27	01
A	0.20	00
D	0.13	00

Algoritmo de Shannon-Fano

C	0.40	1		
B	0.27		1	
A	0.20	0	0	0.33
D	0.13			

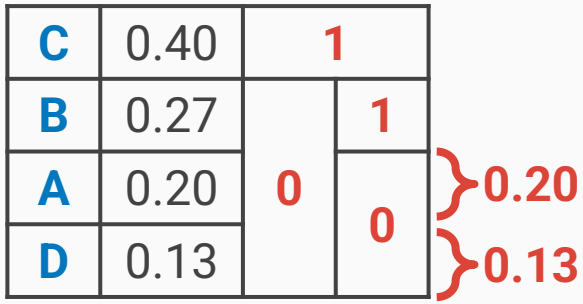
C	0.40	1
B	0.27	01
A	0.20	00
D	0.13	00

Algoritmo de Shannon-Fano



C	0.40	1
B	0.27	01
A	0.20	00
D	0.13	00

Algoritmo de Shannon-Fano



C	0.40	1
B	0.27	01
A	0.20	00
D	0.13	00

Algoritmo de Shannon-Fano

C	0.40	1		
B	0.27	0	1	
A	0.20		0	1
D	0.13			0

C	0.40	1
B	0.27	01
A	0.20	00
D	0.13	00

Algoritmo de Shannon-Fano

<table>
<tr><td>C</td><td>0.40</td><td colspan="3">1</td></tr>
<tr><td>B</td><td>0.27</td><td rowspan="3">0</td><td colspan="2">1</td></tr>
<tr><td>A</td><td>0.20</td><td rowspan="2">0</td><td>1</td></tr>
<tr><td>D</td><td>0.13</td><td>0</td></tr>
</table>

C	0.40	1
B	0.27	01
A	0.20	001
D	0.13	000

Algoritmo de Shannon-Fano

C	0.40	1		
B	0.27	0	1	
A	0.20		0	1
D	0.13			0

C	0.40	1
B	0.27	01
A	0.20	001
D	0.13	000

Algoritmo de Shannon-Fano

C	0.40	1		
B	0.27	0	1	
A	0.20		0	1
D	0.13			0

C	0.40	1
B	0.27	01
A	0.20	001
D	0.13	000

Algoritmo de Shannon-Fano

C	0.40	1		
B	0.27	0	1	
A	0.20		0	1
D	0.13			0

A	0.20	001
B	0.27	01
C	0.40	1
D	0.13	000

Algoritmo de Shannon-Fano

A	0.20	001
B	0.27	01
C	0.40	1
D	0.13	000

$H(S) = 1.89$ bits

$L = 1.93$ bits

$\eta = 0.9772$

$R = 0.0228$

Algoritmo de Shannon-Fano

C A D B C C B A C A B C C D B

1 001 000 01 1 1 01 001 1 001 01 1 1 000 01

A	0.20	**001**
B	0.27	**01**
C	0.40	**1**
D	0.13	**000**

Implementación en Python

```
def algoritmo(probs):
    items = [[p, ""] for p in probs]
    codificar(sorted(items))
    return [code for _, code in items]
```

Codificación de fuentes de información

Teoría de la Información y la Comunicación